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United States Patent Application Publication	20250283689
Kind Code	A1
Publication Date	September 11, 2025
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HANDGUARD MOUNTING SYSTEM

Abstract

A handguard mounting system including a receiver having a tenon on which is defined a plurality of tenon lugs and a handguard having an open rear end in which the tenon is receivable. The handguard includes a plurality of handguard lugs on an interior surface of the open rear end. The tenon lugs and the handguard lugs cooperate to mount the handguard to the receiver when the tenon is received in the open rear end of the handguard and a top of the handguard is rotationally aligned around the tenon with a top of the receiver. One of the handguard or the receiver can include a latch configured to be selectably engaged with the other one of the handguard or the receiver to deter rotation of the handguard about the tenon when the top of the handguard is rotationally aligned around the tenon with the top of the receiver.

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Family ID: **1000008516032**

Appl. No.: **19/074705**

Filed: **March 10, 2025**

Related U.S. Application Data

us-provisional-application US 63563349 20240309

Publication Classification

Int. Cl.: **F41C23/16** (20060101)

U.S. Cl.:

CPC **F41C23/16** (20130101);

Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS [0001] This non-provisional patent application claims priority to U.S. Provisional Patent Application Ser. No. 63/563,349, filed Mar. 9, 2024 and titled “HANDGUARD MOUNTING SYSTEM”, the entire disclosure of which is hereby incorporated by reference.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0003] Not Applicable.

REFERENCE TO SEQUENCE LISTING OR COMPUTER PROGRAM LISTING APPENDIX

[0004] Not Applicable.

BACKGROUND OF THE INVENTION

[0005] The invention relates generally to guns having a handguard. More particularly, the invention pertains to guns having a handguard that surrounds a barrel and is mounted to a receiver of the gun. Known handguard mounting interfaces for mounting a handguard onto a receiver of a gun (including but not limited to air guns and firearms in rifle, pistol, and shotgun configurations) typically require the use of fasteners that extend through the handguard to either screw or clamp the handguard onto a portion of the receiver. The usage of fasteners can be time-consuming and cumbersome, especially when field stripping the gun, because fasteners must be secured using special tools and specific torque requirements. Fasteners can also decrease the durability and the service life of the interface because fasteners inherently create unwanted stress points. Additional problems, including decreased reliability of the gun's sights, may also result from strained or otherwise improperly secured fasteners. Accordingly, what is needed are improvements in handguard mounting systems for guns.

BRIEF SUMMARY

[0006] This Brief Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter. Features of the presently disclosed invention overcome or minimize some or all of the identified deficiencies of the prior art, as will become evident to those of ordinary skill in the art after a study of the information presented in this document.

[0007] Aspects of the disclosure provide a handguard mounting system for a gun. The handguard mounting system includes a receiver with a tenon that has a plurality of tenon lugs extending outwardly therefrom. The handguard mounting system also includes a handguard with an open rear end thereof in which the tenon is receivable. The handguard includes a plurality of handguard lugs on an interior surface of the open rear end. The tenon lugs and the handguard lugs cooperate to prevent longitudinal movement of the handguard relative to the receiver when the tenon is received in the open rear end of the handguard and a top of the handguard is rotationally aligned around the tenon with a top of the receiver. Upon rotating and longitudinally moving the handguard relative to the receiver, the corresponding lugs can selectably and mechanically lock or unlock the handguard to the receiver.

[0008] In some embodiments, cooperation of the tenon lugs and the handguard lugs, and thus rotational alignment of the top of the receiver with the top of the handguard, can be maintained by a latch. The latch can be movably mounted to one of the handguard or the receiver and configured to be selectably engaged with the other one of the handguard or the receiver to deter rotation of the handguard about the tenon when the top of the handguard is rotationally aligned around the tenon with the top of the receiver.

[0009] In one aspect, a handguard mounting system includes a receiver having a tenon on which is defined a plurality of tenon lugs and a handguard having an open rear end in which the tenon is receivable. The handguard includes a plurality of handguard lugs on an interior surface of the open rear end. The tenon lugs and the handguard lugs cooperate to mount the handguard to the receiver when the tenon is received in the open rear end of the handguard and a top of the handguard is rotationally aligned around the tenon with a top of the receiver.

[0010] In another aspect, a method of mounting a handguard for a gun to a receiver for the gun includes an initial step of providing a receiver with lugs and a handguard with lugs. The method further includes the steps of inserting the tenon into the open rear end of the handguard and rotationally aligning the top of the handguard around the tenon with the top of the receiver.

[0011] In another aspect, a handguard mounting system includes a receiver having a tenon on which is defined a plurality of tenon lugs and a handguard having an open rear end in which the tenon is receivable. The handguard includes a plurality of handguard lugs on an interior surface of the open rear end. The tenon lugs and the handguard lugs cooperate to prevent longitudinal movement of the handguard relative to the receiver when the tenon is received in the open rear end of the handguard and a top of the handguard is rotationally aligned around the tenon with a top of the receiver. The handguard mounting system further includes a latch on one of the handguard or the receiver configured to be selectably engaged with the other one of the handguard or the receiver to deter rotation of the handguard about the tenon when the top of the handguard is rotationally aligned around the tenon with the top of the receiver.

[0012] In another aspect, a method of mounting a handguard for a gun to a receiver for the gun includes an initial step of providing a handguard with lugs and a receiver with lugs. The method further includes inserting the tenon into the open rear end of the handguard and rotationally aligning the top of the handguard around the tenon with the top of the receiver. The method further includes a step of engaging the latch with the other one of the handguard or the receiver.

[0013] Numerous other objects, advantages and features of the present disclosure will be readily apparent to those of skill in the art upon a review of the following drawings and description of exemplary embodiments.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0014] Non-limiting and non-exhaustive embodiments are described with reference to the following figures, wherein like reference numerals refer to like parts throughout the various drawings unless otherwise specified. In the drawings, not all reference numbers are included in each drawing, for the sake of clarity.

[0015] FIG. **1** is a perspective side cutaway view of a PRIOR ART handguard mounting interface wherein the handguard is mounted onto a barrel nut that is threaded onto a receiver.

[0016] FIG. **2** is a perspective side view of another PRIOR ART handguard mounting interface wherein the handguard is clamped onto the receiver via fasteners.

[0017] FIG. **3** is a perspective side view of another PRIOR ART handguard mounting interface wherein the handguard is secured directly to the body of the receiver via fasteners.

[0018] FIG. **4** is an elevated front right side perspective view of a gun in the form of a gas operated AR platform rifle equipped with a handguard mounting system constructed in accordance with an embodiment of the present invention.

[0019] FIG. **5** is an elevated front right side perspective view of a front end of the receiver of the handguard mounting system of FIG. **4** showing a plurality of tenon lugs circumferentially disposed about the tenon at front end of the receiver.

[0020] FIG. **6** is an elevated rear right side perspective view of an open rear end of the handguard of the handguard mounting system of FIG. **4** showing a plurality of handguard lugs on the interior surface of the open rear end.

[0021] FIG. **7** is an elevated right side perspective detail view of the handguard mounting system of FIG. **4** showing the handguard securely mounted to the receiver with the top of the handguard rotationally aligned about the tenon with the top of the receiver and latch in the locked position.

[0022] FIG. **8** is a sectional view of the objects of FIG. **7** taken along line **8-8**. The handguard lugs cooperate with and mechanically engage the tenon lugs to prevent the handguard from being longitudinally displaced along the tenon and thereby separated from the receiver.

[0023] FIG. **9** is a right side longitudinal sectional view of the objects of FIG. **7**.

[0024] FIG. **10** is an elevated right side perspective detail view of the handguard mounting system of FIG. **4** showing the latch in the unlocked position and the handguard rotated about the tenon to an intermediate position wherein the handguard lugs do not cooperate with or mechanically engage the tenon lugs to prevent the handguard from being longitudinally displaced along the tenon and thereby removed from the receiver.

[0025] FIG. **11** is a sectional view of the objects of FIG. **10** taken along line **11-11** showing the handguard lugs aligned with a plurality of handguard lug keyways and the tenon lugs aligned with a plurality of tenon lug keyways so that the handguard is removable from the receiver upon forward longitudinal translation of the handguard along the tenon.

[0026] FIG. **12** is an elevated right side perspective longitudinal sectional view of the objects of FIG. **10**.

[0027] FIG. **13** is an elevated rear right side perspective detail view of the objects of FIG. **10** showing the handguard translated longitudinally forward along the tenon for separation from the receiver.

[0028] FIG. **14** is an elevated front right side perspective detail view of the objects of FIG. **13** showing the handguard translated longitudinally further forward along the tenon and separated from the receiver.

[0029] FIG. **15** is an elevated front right side perspective view of a gun in the form of a shotgun equipped with a handguard mounting system constructed in accordance with an embodiment of the present invention.

[0030] FIG. **16** is an elevated front right side perspective detail view of the handguard mounting system of FIG. **15** showing the handguard securely mounted to the receiver with the top of the handguard rotationally aligned about the tenon with the top of the receiver and latch in the locked position. The charging handle is omitted for clarity.

[0031] FIG. **17** is a partial cutaway view of the objects of FIG. **16**.

[0032] FIG. **18** is an elevated front right side perspective view of the handguard mounting system of FIG. **15** showing the latch in the unlocked position and the handguard translated longitudinally forward along the tenon and disconnected from the receiver.

DETAILED DESCRIPTION

[0033] The details of one or more embodiments of the disclosure are set forth in this document. Modifications to embodiments described in this document, and other embodiments, will be evident to those of ordinary skill in the art after a study of the information provided herein. The information provided in this document, and particularly the specific details of the described exemplary embodiment(s), is provided primarily for clearness of understanding and no unnecessary limitations are to be understood therefrom. In case of conflict, the specification of this document, including definitions, will control.

[0034] While the making and using of various embodiments of the disclosure are discussed in detail below, it should be appreciated that the disclosure provides many applicable inventive concepts that are embodied in a wide variety of specific contexts. The specific embodiments discussed herein are merely illustrative of specific ways to make and use the invention and do not delimit the scope of the invention. Those of ordinary skill in the art will recognize numerous equivalents to the specific apparatus and methods described herein. Such equivalents are considered to be within the scope of this invention and are covered by the claims.

[0035] While the terms used herein are believed to be well understood by one of ordinary skill in the art, a number of terms are defined below to facilitate the understanding of the embodiments described herein. Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the subject matter disclosed herein belongs. The terms defined herein have meanings as commonly understood by a person of ordinary skill in the areas relevant to the disclosure. Terms such as “a,” “an,” and “the” are not intended to refer to only a singular entity, but rather include the general class of which a specific example may be used for illustration. The terminology herein is used to describe specific embodiments of the invention, but their usage does not delimit the invention, except as set forth in the claims.

[0036] As described herein, an “upright” position is considered to be the position of apparatus components while in proper operation or in a natural resting position as described and shown herein, for example, in FIGS. **4** and **15**. The upright firing position of a firearm is a generally level firing position. “Vertical,” “horizontal,” “above,” “below,” “side,” “top,” “bottom,” “upper,” “lower,” and other orientation terms are described with respect to this upright position during operation, unless otherwise specified, and are used to provide an orientation of embodiments of the invention to allow for proper description of example embodiments. A person of skill in the art will recognize, however, that the apparatus can assume different orientations when in use. The term “longitudinally spaced” can mean axially spaced along a longitudinal axis of a gun.

[0037] As used herein, the terms “front” and “forward” means in a direction extending toward the muzzle of the gun. In some cases, the term “forward” can also mean forward beyond the muzzle of the gun. The terms “aft” and “rear” means in a direction extending away from the muzzle of the gun toward a rear end of a gun. In some cases, the term “rearward” can also mean rearward beyond the rear end of the gun.

[0038] The term “when” is used to specify orientation for relative positions of components, not as a temporal limitation of the claims or apparatus described and claimed herein unless otherwise specified.

[0039] The terms “above”, “below”, “over”, and “under” mean “having an elevation or vertical height greater or lesser than” and are not intended to imply that one object or component is directly over or under another object or component.

[0040] The phrase “in one embodiment,” as used herein does not necessarily refer to the same embodiment, although it may. Conditional language used herein, such as, among others, “can,” “might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or states. Thus, such conditional language is not generally intended to imply that features, elements and/or states are in any way required for one or more embodiments.

[0041] All measurements should be understood as being modified by the term “about” regardless of whether the word “about” precedes a given measurement.

[0042] The terms “significantly”, “substantially”, “approximately”, “about”, “relatively,” or other such similar terms that may be used throughout this disclosure, including the claims, are used to describe and account for small fluctuations, such as due to variations in manufacturing or processing from a reference or parameter, whether the reference or parameter is explicitly specified or is generally considered normal or possible within the limits of applicable industry-accepted manufacturing practices and tolerances. Such small fluctuations include a zero fluctuation from the reference or parameter as well. For example, they can refer to less than or equal to ±10%, such as less than or equal to ±5%, such as less than or equal to ±2%, such as less than or equal to ±1%, such as less than or equal to ±0.5%, such as less than or equal to ±0.2%, such as less than or equal to ±0.1%, such as less than or equal to ±0.05%.

[0043] The terms “coupled” and “connected” mean at least either a direct electrical or mechanical connection between the connected items or an indirect connection through one or more passive or active intermediary devices.

[0044] All references to singular characteristics or limitations of the present disclosure shall include the corresponding plural characteristic(s) or limitation(s) and vice versa, unless otherwise specified or clearly implied to the contrary by the context in which the reference is made.

[0045] All combinations of method or process steps as used herein can be performed in any order, unless otherwise specified or clearly implied to the contrary by the context in which the referenced combination is made.

[0046] The methods and devices disclosed herein, including components thereof, can comprise, consist of, or consist essentially of the essential elements and limitations of the embodiments described herein, as well as any additional or optional components or limitations described herein or otherwise useful.

[0047] Referring to FIGS. **1-3**, there is shown various prior art handguard mounting interfaces **10** that require the usage of fasteners **12** to mount a handguard **14** onto a receiver **16** (e.g., an upper receiver) of a gun **18**. FIG. **1** illustrates a free-floating handguard mounting interface **10** wherein a handguard (not shown) is fastened directly onto a barrel nut **20** via fasteners **12** which thread into threaded fastener holes **22** in the barrel nut **20**. The barrel nut **20** is threaded onto a threaded tenon **24** of the receiver **16**. The attachment point between the barrel nut **20** and the receiver **16** is a relatively weak attachment point because only a small portion of the receiver **16** supports the barrel nut **20**. Hence, upon attaching the handguard to the barrel nut **20** via the fasteners **12**, the handguard is also poorly supported. FIG. **2** illustrates a clamp-on prior art handguard mounting interface **10** wherein the barrel nut **20** does not include fastener holes **22** and the handguard **14** is clamped onto the receiver **16** via fasteners **12** at the bottom of the handguard **14**. FIG. **3** illustrates a prior art handguard mounting interface **10** wherein the handguard **14** is fastened directly onto the body of the receiver **16** via fasteners **12** that extend through the handguard **14** and screw into corresponding holes within the receiver body. Due to the required use of fasteners **12** to mount the handguard **14**, such prior art handguard mounting interfaces **10** create stress points (at the fasteners **12**) that hinder durability and the service life of the interface **10**, require specific torque requirements and specific tools, such as torque wrenches, require an adhesive, such as thread-lockers (making field stripping or servicing much more difficult), and cause secondary issues such as hampering the gun's sights because of the inherent lack of rigidity in securing fasteners **12**. Additionally, especially for higher caliber platforms, the handguards may become loose due to the strained or improperly secured fasteners **12**.

[0048] Referring to FIGS. **4-14**, there is shown a gun in the form of a gas operated AR platform rifle equipped with a handguard mounting system **100** constructed in accordance with an embodiment of the present invention. The handguard mounting system **100** includes a receiver **102** having a body **130** and a tenon **104** on which is defined a plurality of tenon lugs **106**, and a handguard **108** with an open rear end **110** having a corresponding plurality of handguard lugs **112** defined on an interior surface **114** thereof. The tenon **104** extends forwardly from a forward end of the receiver body **130** and defines a longitudinal axis **136** of the receiver **102**. The tenon **104** is receivable in the open rear end **110** of the handguard **108**. The handguard **108** is received against the forward end of the receiver body **130** when the tenon **104** is received in the open rear end **110** of the handguard **108**. When the tenon **102** is received in the open rear end **110** of the handguard **108** and a top **142** of the handguard **108** is rotationally aligned about the tenon **104** with a top **132** of the receiver **102**, the handguard lugs **112** cooperate with (i.e., mechanically engage, as explained

in more detail below) the tenon lugs **106** to prevent longitudinal movement of the handguard **108** relative to the receiver **102** and thereby mount the handguard **108** to the receiver **102**. A latch **160** on the receiver **102** is then selectably actuatable to engage a portion of the handguard **108** and thereby secure the handguard **108** to the receiver **102** by deterring rotation of the handguard **108** about the tenon **104**. It is to be understood, however, that although the latch **160** is depicted in the figures as being mounted to the receiver **102** and configured to selectably engage a portion of the handguard **108**, in some embodiments, the latch **160** can alternatively be mounted to the handguard **108** and configured to selectably engage a portion of the receiver **102**.
[0049] The handguard mounting system **100** is essentially a locking mechanism for mounting the handguard **108** to the receiver **102** of a gun **116**. Importantly, the handguard mounting system **100** is usable with any desired gun **116** or other weapon that uses a handguard **108**. For example, in some embodiments, the gun **116** can be an air gun or a firearm configured as a rifle, a pistol, or a shotgun. In some embodiments, the gun **116** can be a grenade launcher, a less-lethal projectile launcher, or any other projectile launching weapon having a handguard that mounts to a receiver. As shown in FIGS. **4-14**, the gun **116** is a gas operated AR platform rifle with a barrel **118** extending through the tenon **104**, a gas tube **120**, and a gas block (not shown). A top portion of the tenon **104** includes a groove **122** (FIG. **5**) configured to receive a gas tube **120** extending the full length thereof, and the handguard **108** includes a widened groove **124** at the top of the open rear end **110** (FIG. **6**) configured to accommodate the gas tube **120** during mounting and dismounting of the handguard **108**. In some embodiments, the gun **116** is not a gas operated rifle and grooves **122**, **124** can be omitted.
[0050] In comparison to prior art handguard mounting interfaces **10**, which require fasteners **12** and mounting bracketry, the novel handguard mounting system **100** of the instant invention reduces the number of parts that are required to securely mount the handguard **108** to the receiver **102**. This makes the gun **116** more easily field strippable and makes the handguard **108** readily removable without the need of tools (e.g., Allen wrench, torque wrench). The omission of fasteners (e.g., screws) also beneficially reduces stress points. The handguard mounting system **100** also provides significantly more contact surface area between the handguard **108** and the receiver **102**, which yields a more robust construction that increases rigidity, durability, and overall strength of the attachment therebetween. The area of attachment includes nearly the full outer surface area of the tenon **104**, which mates with the interior surface **114** of the open rear end **110** of handguard **108**, as well as certain surfaces of the complimentary tenon and handguard lugs **106**, **112**, respectively, which mechanically engage one another to prevent longitudinal movement of the handguard **108** away from the receiver body **130**. This better maintains the concentricity of the handguard **108** with the barrel **118** (FIG. **9**) and increases reliability and repeatability of handguard attachment, which in turn provides improvements in consistency of sight accuracy (e.g., front-end sights, such as iron sights, lasers, or other optics) and shot placement.
[0051] Referring to FIG. **5**, the receiver **102** includes a receiver body **130** having a top **132** and a bottom **134** opposite the top **132**. The tenon **104** protrudes forwardly from a forward end of the receiver body **130**. The tenon **104** defines a longitudinal axis **136** of the receiver **102**. The tenon **104** can be integrally formed with the receiver body **130**. The tenon lugs **106** are spaced circumferentially around and extend radially outward from an outer surface **138** of the tenon **104**.
[0052] Referring to FIG. **6**, the handguard **108** includes a handguard body **140** having a top **142** and a bottom **144** opposite the top **142**. In one example, the top **132** of the receiver **102** and the top **142** of the handguard **108** define a continuous rail **132**, **142**, such as a picatinny rail. In rotationally aligning the top **142** of the handguard **108** with the top **132** of the receiver **102**, the handguard **108** is rotationally indexed to the receiver **102** (e.g., wherein the tops **132**, **142** rest substantially flush with one another and form a continuous rail). The handguard lugs **112** are spaced circumferentially around and protrude centripetally (i.e., inwardly) from the interior surface **114** of the open rear end **110** of the handguard **108**. The tenon **104** is receivable in the handguard **108** through the open rear

end **110**. The handguard body **140** is received against the forward end of the receiver body **130** when the tenon **104** is received in the open rear end **110** of the handguard **108**.
[0053] In one embodiment, the lugs of the receiver **102** are circumferentially spaced apart from one another at an arcuate distance about the circumference of the tenon **104**, thus defining handguard lug keyways **146** (or gaps or channels) that extend longitudinally along the tenon **104** between adjacent tenon lugs **106**. The handguard lug keyways **146** are sized to receive the handguard lugs **112** therethrough. The handguard lugs **112** are likewise circumferentially spaced apart from one another at an arcuate distance about the circumference of the interior surface **114** of the open rear end **110** of the handguard **108** so as to define tenon lug keyways **148** (or gaps or channels) extending longitudinally along the interior surface **114** of the open rear end **110** of the handguard **108** between adjacent handguard lugs **112**. The tenon lug keyways **148** are sized to receive the tenon lugs **106** therethrough. The tenon and handguard lugs **106**, **112**, and handguard and tenon lug keyways **146**, **148** are complementary to one another in that the handguard lugs **112** can pass through the handguard lug keyways **146** on the tenon **104**, and the tenon lugs **106** can pass through the tenon lug keyways **148** on the handguard **108** when the tenon **104** is inserted into or removed from the open rear end **110** of the handguard **108** during assembly or disassembly of the mounting system **100**. As such, when the tenon and handguard lugs **106**, **112** are aligned with the respective handguard and tenon lug keyways **146**, **148**, the handguard **108** is free to translate longitudinally along the tenon **104** (and the longitudinal axis **136**) onto or off of the receiver **102**. However, when the tenon **102** is received in the open rear end **110** of the handguard **108** and the top **142** of the handguard **108** is rotationally aligned about the tenon **104** with the top **132** of the receiver **102**, a forward surface **172** of each handguard lug **112** mechanically engages a rear surface **174** of each tenon lug **106** to prevent the tenon **104** from being removed from the open rear end **110** of the handguard **108**. In this way, the handguard lugs **112** longitudinally overlap the tenon lugs **106**. The tenon **104** and handguard **108** also define circumferential (i.e., annular) gaps or channels **150**, **152** around the outer surface of the tenon **104** and the interior surface **114** of the open rear end **110** of the handguard **108**, respectively, through which the corresponding handguard lugs **112** and tenon lugs **106** may pass (once the handguard **108** is received against the front end of the tenon body **130**) during rotation of the handguard **108** about the tenon **104** to align the top **142** of the handguard **108** with the top **132** of the receiver **102**.
[0054] In one embodiment, the handguard mounting system **100** further includes a selectably lockable latch **160** configured to rotationally lock the handguard **108** relative to the receiver **102**. The latch **160** is pivotably mounted in a mounting slot or recess **162** defined in the receiver body **130** proximate the tenon **104**. The latch **160** is selectably receivable within a corresponding slot or recess **164** in the body of the handguard **108** to rotationally lock the handguard **108** relative to the receiver **102** and thus prevent the handguard **108** from rotating about the tenon **104** of the receiver **102**. The latch **160** is thus movable between a locked position (see e.g., FIGS. **7** and **9**) wherein the latch **160** is received within a slot or recess **164** in the body of the handguard **108**, and an unlocked position (see e.g., FIG. **12**) wherein the latch **160** is not received within the slot or recess **164** in the body of the handguard **108**.
[0055] When in the locked position, the latch **160** deters or prevents the handguard **108** from rotating about the tenon **104** relative to the receiver **102** and, in combination with the cooperating tenon lugs **106** and handguard lugs **112** that simultaneously prevent longitudinal displacement of the handguard **108** relative to the receiver **102**, functions to index and rigidly secure or lock the handguard **108** to the receiver **102**. As such, the latch **160** can be said to function like a cross bar by preventing rotation of the handguard **108** about the tenon **104** when in the locked position. When in the unlocked position, the latch **160** does not deter or prevent the handguard **108** from rotating about the tenon **104** relative to the receiver **102**, which in turn permits the handguard **108** to rotate freely about the tenon **104** of the receiver **102** so that the handguard **108** can be mounted on or removed from the tenon **104**.

[0056] In one embodiment, the latch **160** takes the form of a pivotable lever **160**. The lever **160** includes a lever body **166** that is pivotably mounted within the slot **162** of the receiver body **130** about a pivot pin (unnumbered). The lever **160** further includes a spur **168** protruding from the lever body **166** which is shaped and sized to be receivable within the slot **164** of the handguard body **140** to deter rotation of the handguard **108** about the tenon **104** when the latch **160** is in the locked position. The latch **160** is configured to remain in the locked position (i.e., maintain the spur **168** in the slot **164**) until the latch **160** is manually moved to the unlocked position. In one embodiment, the lever body **166** can define a camming surface **170** configured to maintain the spur **168** in the slot **164** when the lever **160** is in the locked position until the spur **168** is manually lifted out of the slot **164** to move the lever **160** to the unlocked position. The camming surface **170** can be an arcuate surface that engages a bottom wall of the slot **162** in which the lever body **166** is pivotably mounted when the spur **168** is received in the slot **164** of the handguard (i.e., in the locked position). The latch **160** is manually movable by the user between the locked and unlock positions. As used herein, the term "manually" can include operation by hand or using a hand tool. Put differently, the user can move the latch **160** between the locked and unlock positions by hand or using a hand tool such as a screw driver, blade, or other narrow implement. In one embodiment, when in the locked position, the latch **160** can rest completely within the aligned slots **162**, **164** in the receiver body **130** and the handguard body **140**. This protects the latch **160** from becoming inadvertently bumped or knocked out of the locked position and into the unlocked position.

[0057] As noted above, it is to be understood that although the latch **160** is depicted in FIGS. **4-18** as being pivotably mounted to the receiver **102** and configured to selectably engage a slot **164** in the handguard **108**, in other embodiments, the latch **160** can alternatively be pivotably mounted to the slot **164** in the handguard **108** and configured to selectably engage the groove or slot **162** in the receiver body **130**. It should also be understood that the possible forms of the locking latch disclosed herein need not mirror that of the lever **160** depicted in the figures. For example, in some embodiments not specifically illustrated herein, the latch **160** can alternatively take the form of a manually actuatable plunger that is mounted to one or the other of the receiver **102** or the handguard **108**. In such embodiments, a retractable portion of the plunger is receivable in a corresponding hole or depression defined in the other one of the receiver **102** or the handguard **108**. Additional latching mechanisms will be apparent to those of ordinary skill in the art. All such latching mechanisms are considered within the scope of the invention as set forth in the claims.

[0058] FIGS. **7-9** illustrate a locked or working position of the handguard mounting system **100**, wherein the tenon **104** is fully seated within the open rear end **110** of the handguard **108**, the tops **132**, **142** of the receiver **102** and handguard **108** are rationally aligned (FIG. **7**), and the latch **160** is engaged to prevent the handguard **108** from rotating relative to the receiver **102** (FIG. **9**). In the locked position, the handguard **108** is received against the forward end of the receiver body **130** when the tenon **104** is received in the open rear end **110** of the handguard **108** (such that the handguard **108** butts against the receiver body **130**). As shown in FIG. **8**, in the locked position, the lugs **106**, **112** are misaligned from the keyways **146**, **148** and engaged with one another, preventing the handguard **108** from moving longitudinally relative to the receiver **102** (e.g., moving linearly forward along the longitudinal axis **136** and separating from the receiver **102**). In more detail, the engagement of the tenon lugs **106** and the handguard lugs **112** prevents longitudinal movement of the handguard **108** relative to the receiver **102** when the handguard **108** is received against the receiver **102**, with the tenon **104** received in the open rear end **110** of the handguard **108**, and the top **142** of the handguard **108** is rotationally aligned with the top **132** of the receiver **102**. Thereby, the lugs **106**, **112** cooperate (e.g., engage) to prevent the handguard **108** from moving longitudinally off the tenon **104**, and the latch **160** prevents the handguard **108** from rotating to a position where the lugs **106**, **112** are disengaged to allow the handguard **108** to be pulled off the tenon **104**. Said differently, the latch **160** in tandem with the lugs **106**, **112**, rotationally and linearly fix the handguard **108** relative to the receiver **102**, because the latch **160** in the locked position, prevents any rotational movement of the handguard **108** which in turn maintains lug engagement (i.e., preventing a rotation necessary for lug disengagement or lug-keyway alignment). In the locked position, the handguard body **140** of the handguard **108** completely covers and protects the lugs **106**, **112**.

[0059] In some embodiments, one or both of the pluralities of tenon lugs **106** and handguard lugs **112** can be configured to bias the handguard **108** into contact with the receiver body **130** when the tenon **104** is received in the open rear end **110** of the handguard **108** and the top **142** of the handguard **108** is rotationally aligned around the tenon **104** with the top **132** of the receiver **102**. For example, in some embodiments, one or both of the pluralities of tenon lugs **106** and handguard lugs **112** can be angled relative to the longitudinal axis **136** of the tenon **104**. In embodiment, at least one plurality of the plurality of tenon lugs **106** and the plurality of handguard lugs **112** can be at least partially helical relative to the longitudinal axis of the tenon **104**. Consequently, when the tenon **104** is received in the open rear end **110** of the handguard **108** and the top **142** of the handguard **108** is rotationally aligned around the tenon **104** with the top **132** of the receiver **102**, cooperation of the helical tenon lugs **106** and/or handguard lugs **112** can bias or translate the handguard rearwardly along the tenon **104** like screw threads.

[0060] In embodiments wherein one or both of the pluralities of tenon lugs **106** and handguard lugs **112** are angled or helical, the handguard **108** can be rotatable about the tenon **104** in only one of a clockwise direction or a counterclockwise direction to rotationally align the top **142** of the handguard **108** with the top **132** of the receiver **102** and thus mount the handguard **108** to the receiver **102**. In other embodiments wherein the pluralities of tenon lugs **106** and handguard lugs **112** are not angled or helical, the handguard **108** can be rotatable about the tenon **104** in either of a clockwise direction and a counterclockwise direction to rotationally align the top **142** of the handguard **108** with the top **132** of the receiver **102** and thus mount the handguard **108** to the receiver **102**.

[0061] In one embodiment, the tenon lugs **106** of the receiver **102** and the handguard lugs **112** of the handguard **108** include multiple circumferential rows of lugs **180**, **182**, **190**, **192**, respectively (FIGS. **5** and **6**). For example, the tenon **104** includes a first row **180** and a second row **182** of tenon lugs **106** spaced longitudinally rearward on the tenon **104** from the first row **180** of tenon lugs **106**. Similarly, the handguard **108** includes a first row **190** and a second row **192** of handguard lugs **112** spaced longitudinally forward on the interior surface **114** from the first row **190** of handguard lugs **112**. The space or distance between rows **180**, **182**, **190**, **192** of lugs **106**, **112** (which defines the circumferential channels **150**, **152**) can be approximately equal to or slightly larger than a width of the corresponding lug **106**, **112** which is receivable in the channel **150**, **152**. In one embodiment, each plurality of tenon lugs **106** and handguard lugs **112** includes only a single row of lugs **106**, **112**. In other embodiments, each plurality of tenon lugs **106** and handguard lugs **112** can include two or more rows lugs **106**, **112**. In one embodiment, the widths of all of the tenon lugs **106** and handguard lugs **112** are approximately equal.

[0062] In one embodiment, each handguard lug **112** is longer than each tenon lug **106**. For example, each handguard lug **112** extends further around the interior surface **114** of the open rear end **110** of the handguard **108** than each tenon lug **106** extends around the outer surface **138** of the tenon **104**. The use of longer handguard lugs **112** increases the contact or mating surface area between the handguard lugs **112** and the outer surface **138** of the tenon **104**, which thereby reduces wear, increases rigidity and durability, and further increases the service life of the handguard mounting system **100**. In alternate embodiments, each tenon lug **106** can be longer than each handguard lug **112**. In some embodiments, the tenon lugs **106** and the handguard lugs **112** can be substantially equal in length. As can be appreciated, the lengths and surface profiles of the lugs **106**, **112** can determine the tolerances between the handguard **108** and the receiver **102**, when linearly moving and/or rotating the handguard **108** relative to the receiver **102**.

[0063] In one embodiment, each tenon lug **106** can have a substantially cylindrical cross section and a tapered at each lateral end. The tapered lateral ends create a seamless intersection with the outer surface **138** of the tenon **104**. In one embodiment, the shape of the handguard lugs **112** can correspond to the shape and size of the outer surface **138** of the tenon **104** and the tenon lugs **106**. In an example, each handguard lug **112** can have concave and tapered lateral ends that correspond to the shape of the tenon lugs **106** and a concave middle section that corresponds to the shape of the outer surface **138** of the tenon **104**.

[0064] FIGS. **10-12** depict the handguard mounting system **100** with the latch **160** in the unlocked position and the handguard **108** rotated about the tenon **104** to an intermediate position wherein the handguard lugs **112** do not cooperate with or mechanically engage the tenon lugs **104** to prevent the handguard **108** from being longitudinally displaced along the tenon **104** and thereby removed from the receiver **102**. When the handguard **108** is in the intermediate position, the tenon and handguard lugs **106**, **112** are aligned with the respective tenon lug and handguard lug keyways **148**, **146**. More specifically, when the handguard **108** is in the intermediate position, the handguard lug keyways **146** on the tenon **104** align with the handguard lugs **112**, and the tenon lug keyways **148** on the handguard **108** align with the tenon lugs **106** of the tenon **104**. This allows the handguard **108** to be longitudinally displaced (i.e., linearly moved) along the longitudinal axis **136** of the tenon **104**, as the latch **160** will be in the unlocked position (i.e., disengaged from the handguard). The intermediate position of the handguard can also thus be referred to as a lug-keyway alignment position. FIGS. **13-14** illustrate an unlocked and fully disconnected position of the handguard mounting system **100**, wherein the handguard **108** is completely disconnected from the receiver **102**.

[0065] In use, to mount the handguard **108** to the receiver **102**, a user inserts the tenon **104** into the open rear end **110** of the handguard **108** while the latch **160** is in the unlocked position. To do this, the handguard lugs **112** and tenon lugs **106** must be respectively aligned with the handguard lug keyways **146** and the tenon lug keyways **148**, as best shown in FIG. **11**. Once the tenon **104** is received in the open rear end **110** of the handguard **108** with the handguard **108** abutting the receiver body **130**, the tenon and handguard lugs **106**, **112** will be automatically received within the respective annular channels or gaps **150**, **152** so that the handguard **108** can be rotated about the tenon **104**. The user then rotationally aligns the tops **132**, **142** of the receiver **102** and the handguard **108** by rotating the handguard **108** about the tenon **104**. Upon aligning the tops **132**, **142** of the receiver **102** and handguard **108**, the corresponding slots **162**, **164** in each will also become aligned and indexed to one another. This allows the latch **160** to be moved from the unlocked to the locked position. As such, the user then moves the latch from the unlocked to the locked position by pivoting the lever **160** such that the spur **168** becomes seated in the corresponding slot (which can be slot **164** or slot **162**, depending on whether the latch **160** is mounted to the receiver **102** or the handguard **108**). The handguard **108** will then be rigidly secured to the receiver **102**. The handguard **108** is removable from the receiver **102** by reverse operation of the above method steps.

[0066] FIGS. **15-18** illustrate another embodiment of a handguard mounting system **100** which is incorporated into a shotgun **116**. The handguard mounting system of FIGS. **15-18** is identical to the handguard mounting system of FIGS. **4-14** in all respects except as specifically described hereafter. The tenon lugs **106** and the handguard lugs **112** used in the handguard mounting system **100** on the shotgun **116** are larger in size and more robust in order to resist the greater recoil impulse generated upon discharge of shotgun ammunition. Additionally, the tenon **104** and handguard **108** used in the handguard mounting system **100** on the shotgun **116** omit the gas tube grooves **122**, **124** found on the handguard mounting system of FIGS. **4-14**.

[0067] Although various embodiments have been described in detail, it will be understood by those skilled in the art that various modifications can be made therein without departing from the spirit and scope of the invention as set forth in the appended claims.

[0068] This written description uses examples to disclose the invention and also to enable any person skilled in the art to practice the invention, including making and using any devices or

systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.
[0069] It will be understood that the particular embodiments described herein are shown by way of illustration and not as limitations of the invention. The principal features of this invention may be employed in various embodiments without departing from the scope of the invention. Those of ordinary skill in the art will recognize numerous equivalents to the specific apparatus and methods described herein. Such equivalents are considered to be within the scope of this invention and are covered by the claims.
[0070] All of the compositions and/or methods disclosed and claimed herein may be made and/or executed without undue experimentation in light of the present disclosure. While the compositions and methods of this invention have been described in terms of the embodiments included herein, it will be apparent to those of ordinary skill in the art that variations may be applied to the compositions and/or methods and in the steps or in the sequence of steps of the method described herein without departing from the concept, spirit, and scope of the invention. All such similar substitutes and modifications apparent to those skilled in the art are deemed to be within the spirit, scope, and concept of the invention as defined by the appended claims.
[0071] Thus, although there have been particular embodiments described herein, it is not intended that such references be construed as limitations upon the scope of this invention except as set forth in the following claims.

Claims

1. A handguard mounting system, comprising: a receiver having a tenon on which is defined a plurality of tenon lugs; and a handguard having an open rear end in which the tenon is receivable and a plurality of handguard lugs on an interior surface of the open rear end, wherein the tenon lugs and the handguard lugs cooperate to mount the handguard to the receiver when the tenon is received in the open rear end of the handguard and a top of the handguard is rotationally aligned around the tenon with a top of the receiver.
2. The handguard mounting system of claim 1, wherein: the handguard lugs and the tenon lugs cooperate to prevent longitudinal movement of the handguard relative to the receiver when the tenon is received in the open rear end of the handguard and the top of the handguard is rotationally aligned around the tenon with the top of the receiver.
3. The handguard mounting system of claim 2, wherein: the handguard lugs longitudinally overlap the tenon lugs when the tenon is received in the open rear end of the handguard and the top of the handguard is rotationally aligned around the tenon with the top of the receiver.
4. The handguard mounting system of claim 2, wherein: a forward surface of each handguard lug engages a rear surface of each tenon lug when the tenon is received in the open rear end of the handguard and the top of the handguard is rotationally aligned around the tenon with the top of the receiver.
5. The handguard mounting system of claim 1, wherein: at least one plurality of the plurality of tenon lugs and the plurality of handguard lugs is configured to bias the handguard into contact with a body of the receiver when the tenon is received in the open rear end of the handguard and the top of the handguard is rotationally aligned around the tenon with the top of the receiver.
6. The handguard mounting system of claim 5, wherein: at least one plurality of the plurality of tenon lugs and the plurality of handguard lugs is at least partially helical.
7. The handguard mounting system of claim 1, wherein: the plurality of tenon lugs includes a first row of tenon lugs and a second row of tenon lugs spaced longitudinally rearward on the tenon from

the first row of tenon lugs; and the plurality of handguard lugs includes a first row of handguard lugs and a second row of handguard lugs spaced longitudinally forward on the interior surface of the open rear end from the first row of handguard lugs.
8. The handguard mounting system of claim 1, wherein: each handguard lug of the plurality of handguard lugs extends further around the interior surface of the open rear end of the handguard than each tenon lug of the plurality of tenon lugs extends around the tenon.
9. The handguard mounting system of claim 1, wherein: the plurality of tenon lugs is spaced circumferentially around and extends radially outward from the tenon; and the plurality of handguard lugs is spaced circumferentially around and protrudes centripetally from the interior surface of the open rear end of the handguard.
10. The handguard mounting system of claim 9, wherein: the plurality of handguard lugs defines a plurality of tenon lug keyways through which the plurality of tenon lugs is receivable extending longitudinally along the interior surface of the open rear end of the handguard between adjacent handguard lugs of the plurality of handguard lugs; and the plurality of tenon lugs defines a plurality of handguard lug keyways through which the plurality of handguard lugs is receivable extending longitudinally along the tenon between adjacent tenon lugs of the plurality of tenon lugs.
11. The handguard mounting system of claim 1, wherein: the handguard is rotatable about the tenon in both a clockwise direction and a counterclockwise direction to rotationally align the top of the handguard with the top of the receiver to mount the handguard to the receiver when the tenon is received in the open rear end of the handguard.
12. The handguard mounting system of claim 1, wherein: the handguard is rotatable about the tenon in only one of a clockwise direction or a counterclockwise direction to rotationally align the top of the handguard with the top of the receiver to mount the handguard to the receiver when the tenon is received in the open rear end of the handguard.
13. The handguard mounting system of claim 1, further comprising: a latch on one of the handguard or the receiver, the latch configured to selectably deter rotation of the handguard about the tenon when the top of the handguard is rotationally aligned around the tenon with the top of the receiver.
14. The handguard mounting system of claim 13, wherein: the latch is movable between a locked position wherein the latch engages the other one of the handguard or the receiver to deter rotation of the handguard about the tenon, and an unlocked position wherein the latch does not engage the other one of the handguard or the receiver to deter rotation of the handguard about the tenon.
15. The handguard mounting system of claim 14, wherein: the latch is configured to remain in the locked position until manually moved to the unlocked position.
16. The handguard mounting system of claim 14, wherein: the latch is a lever having a body and a spur protruding from the body; the body is pivotably mounted to the one of the handguard or the receiver; and the other one of the handguard or the receiver defines a slot in which the spur is receivable to deter rotation of the handguard about the tenon when the latch is in the locked position.
17. The handguard mounting system of claim 16, wherein: the body of the lever defines a camming surface configured to maintain the spur in the slot when the lever is in the locked position until the spur is manually lifted out of the slot to move the lever to the unlocked position.
18. A method of mounting a handguard for a gun to a receiver for the gun, comprising: providing the receiver and the handguard of claim 1; inserting the tenon into the open rear end of the handguard; and rotationally aligning the top of the handguard around the tenon with the top of the receiver.
19. A gun comprising the handguard mounting system of claim 1.
20. A handguard mounting system, comprising: a receiver having a tenon on which is defined a plurality of tenon lugs; a handguard having an open rear end in which the tenon is receivable and a plurality of handguard lugs on an interior surface of the open rear end, wherein the tenon lugs and

the handguard lugs cooperate to prevent longitudinal movement of the handguard relative to the receiver when the tenon is received in the open rear end of the handguard and a top of the handguard is rotationally aligned around the tenon with a top of the receiver; and a latch on one of the handguard or the receiver configured to be selectably engaged with the other one of the handguard or the receiver to deter rotation of the handguard about the tenon when the top of the handguard is rotationally aligned around the tenon with the top of the receiver.

21. A method of mounting a handguard for a gun to a receiver for the gun, comprising: providing the receiver and the handguard of claim **19**; inserting the tenon into the open rear end of the handguard; rotationally aligning the top of the handguard around the tenon with the top of the receiver; and engaging the latch with the other one of the handguard or the receiver.